Algorithmic Legendrian Realization of Open Book Decompositions of Contact 3-manifolds

Open books, ribbons, and explicit Legendrian realization

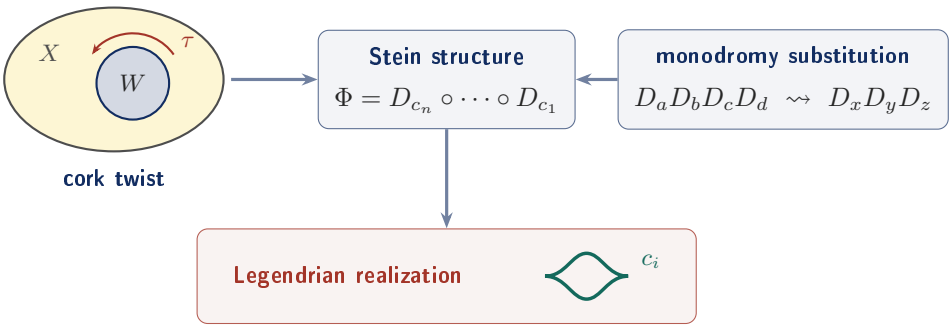
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October 3rd, 2026

Based on R. Avdek, *Algebr. Geom. Topol.* 13 (2013)
and E. Stenhede, arXiv:2604.08010 (2026)

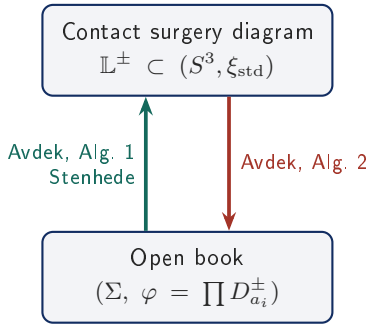
Motivation



[CFHS96; AK12; AY13] [LP01; AO01; KOU17] [EG10; EMV11] [Avd13; Ste26]

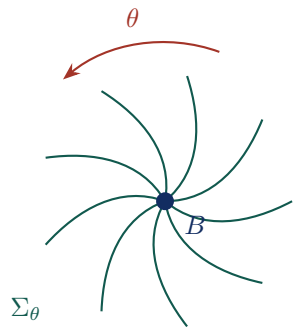
Outline

1. Open books and the Giroux correspondence
2. Contact cell decompositions and ribbons
3. Avdek's algorithms
   surgery diagrams $\rightleftarrows$ open books, with examples
4. Stenhede's improvement
   the algorithm, with an example
5. Computation



Open Books

- **Open book** (B, π) on a closed oriented M^3
  binding B, fibration $\pi\colon M \setminus B \to S^1$, pages Σ
- **Abstract open book** (Σ, φ)
  monodromy $\varphi \in \mathrm{MCG}(\Sigma, \partial\Sigma)$
- **Supports** $\xi = \ker \alpha$
  $\alpha > 0$ on B, $d\alpha > 0$ on every page



[Gir02; Etn06] picture: a standard schematic

The Giroux Correspondence

Theorem (Giroux 2002)

$$\frac{\{\text{contact structures on } M\}}{\text{isotopy}} \quad \longleftrightarrow \quad \frac{\{\text{open books on } M\}}{\text{isotopy, positive stabilization}}$$

M closed, oriented; contact structures positive and cooriented

Surgery = Dehn twist

$K \subset \Sigma$ Legendrian: contact (± 1)-surgery on $K \rightsquigarrow (\Sigma,\ \varphi \circ D_K^{\mp})$

Surgery diagram $\approx$ open book with a Dehn twist factorization

[TW75; Gir02; BHH24; LV24a; LV24b] dictionary: [Gay02; Etn06]

Contact Cell Decompositions and Ribbons

- **Contact cell decomposition**: Legendrian 1–skeleton G, 2–cells with $\mathrm{tb}(\partial D) = -1$, tight 3–cells
- **Ribbon** R_G: thin surface around G, transverse to a Reeb field; convex with dividing set ∂R_G

Theorem (Giroux)

If G is the 1–skeleton of a contact cell decomposition, then R_G is a page of an open book supporting ξ.

Consequence

any Legendrian link extends to such a $G \Rightarrow$ it lies on a page

Legendrian realization

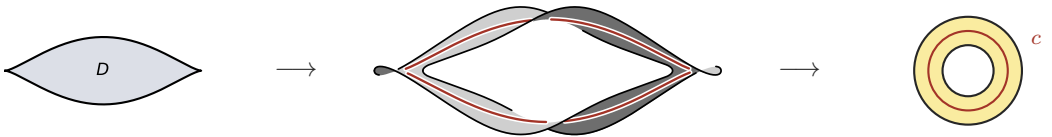
simple closed $\mathfrak{a} \subset R_G$ realizable $\iff [\mathfrak{a}] \neq 0$

existence only — no front

[Gir02; Etn06; Kan98; Hon00]

Model Ribbons (1): the $\mathrm{tb} = -1$ Unknot

One 2–cell D; the ribbon is an annulus whose core is the unknot c



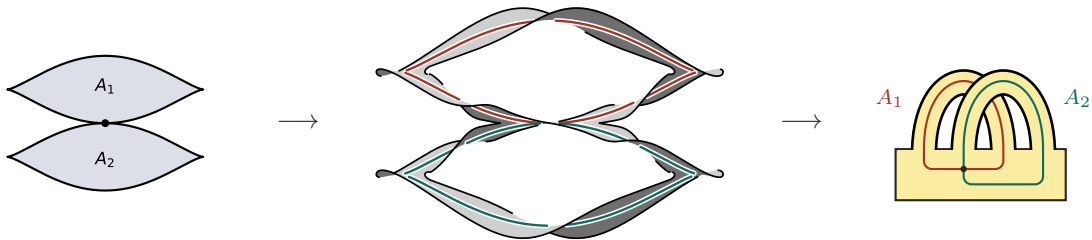
front ribbon (Avdek's rules) abstract page

$(S^1 \times [0,1],\ D_c^+)$ **supports** $(S^3, \xi_{\mathrm{std}})$ (the positive Hopf band)

graph: [Avd13, Fig. 4]; ribbon: [Avd13, Figs. 4, 13–15] page: standard picture

Model Ribbons (2): Two Unknots at a Vertex

Two 2–cells; the ribbon is a punctured torus on which A_1, A_2 meet once, at the vertex



front ribbon (Avdek's rules) abstract page

$(\Sigma_{1,1},\ D^+_{A_2} \circ D^+_{A_1})$ **supports** (S^3, ξ_{std})

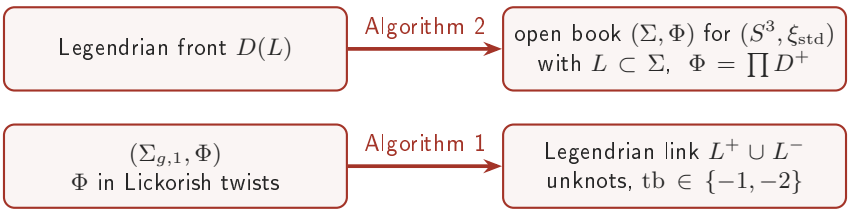
graph: [Ste26, Fig. 6.1], [Avd13, Fig. 7]; ribbon: [Avd13, Figs. 7, 13–15] page: standard picture

Avdek's Algorithms

Earlier: template pages

Akbulut–Ozbagci, Plamenevskaya (also Stipsicz, Arıkan): embed a bridge presentation of L into a template page of an open book for (S^3, ξ_{std})

Avdek: build the page from the front itself

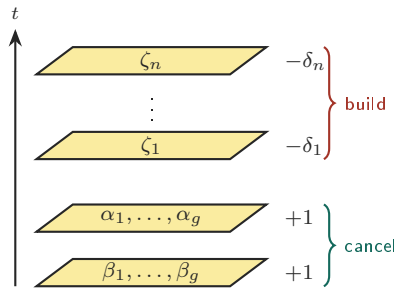


[AO01; Pla04; Sti05; Ari07; Avd13]

Open Books $\to$ Surgery Diagrams

$$(\Sigma_{g,1},\ \Phi = D_{\zeta_n}^{\delta_n} \circ \cdots \circ D_{\zeta_1}^{\delta_1}), \quad \delta_k \in \{+,-\}, \quad \zeta_k \text{ Lickorish generators}$$

0. **Goal**: write $\Phi = \Phi \circ \Phi_{S^3}^{-1} \circ \Phi_{S^3}$
1. **Embed** Σ as a page for (S^3, ξ_{std})
   $\Phi_{S^3} = \prod_i D_{\beta_i} \circ D_{\alpha_i}$
2. **Cancel** Φ_{S^3}
   $(+1)$ on $\alpha_j,\ \beta_j$
3. **Build** Φ

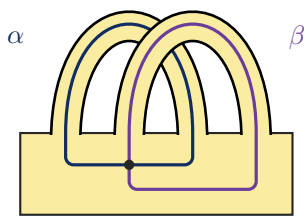


Every closed (M,ξ) is (± 1)–surgery on Legendrian unknots in (S^3, ξ_{std}), $\text{tb} \in \{-1,-2\}$

Avdek, Algorithm 1 [Avd13, §3] refines [DG04]

Example: $L(2,1)$ — Open Book

Input for Algorithm 1: connected binding, monodromy in Lickorish twists



$\Sigma_{1,1}$ with Lickorish curves α, β

Monodromy

$$\Phi = D_\alpha^+ \circ D_\alpha^+ \circ D_\beta^+$$

Avdek's order: $\zeta_1 = \beta,\ \zeta_2 = \zeta_3 = \alpha,$ all $\delta_k = +$

Which manifold?

β crosses a 1–handle once $\Rightarrow$ positive stabilization of (annulus, D_α^2)

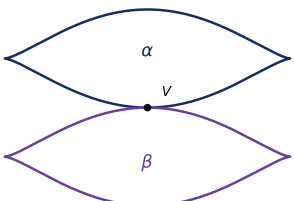
$$\Rightarrow\ L(2,1) = \mathbb{RP}^3,\ \text{tight}$$

[Avd13, §3] picture: standard punctured torus

Example: $L(2,1)$ — Page in (S^3, ξ_{std})

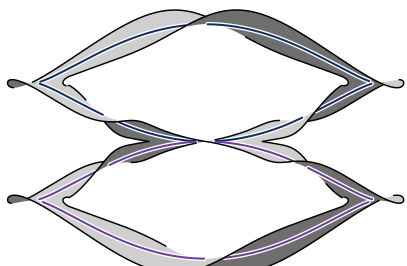
Step 1: $\Sigma_{1,1}$ = ribbon of $\alpha \cup \beta$, a page with $\Phi_{S^3} = D_\beta^+ \circ D_\alpha^+$ (Lemma 3.1)

Legendrian graph



two $\mathrm{tb} = -1$ unknots tangent at V

Page $\Sigma_{1,1}$ with α, β

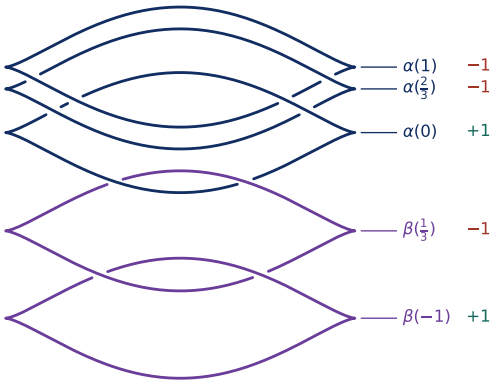


Avdek's local models: cusps, vertex V, one twist per arc

[Avd13, §3, Lemma 3.1, Figs. 7, 13–15] cf. [Ste26, Fig. 6.1]

Example: $L(2,1)$ — Surgery Diagram

$\zeta(t)$ = vertical push-off of ζ by t: distinct heights = distinct pages



Step 2: cancel Φ_{S^3}

$\beta(-1)$, $\alpha(0)$: contact $+1$

Steps 3–5: build Φ

$\beta(\frac{1}{3})$, $\alpha(\frac{2}{3})$, $\alpha(1)$: contact -1

$\Rightarrow \quad (\mathbb{RP}^3,\ \xi_{\text{tight}})$

Algorithm 1 [Avd13, §3, Prop. 3.2] layout as in [Avd13, Fig. 9]

Stenhede's Improvement

Limitation of Algorithm 1

only Lickorish curves on $\Sigma_{g,1}$ — arbitrary curves need an *explicit* Legendrian realization



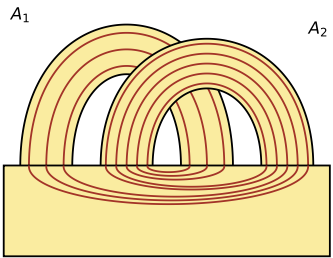
Uniqueness (Thm. 4.2)

Any two Legendrian realizations of the same isotopy class of curves on a ribbon are Legendrian isotopic.

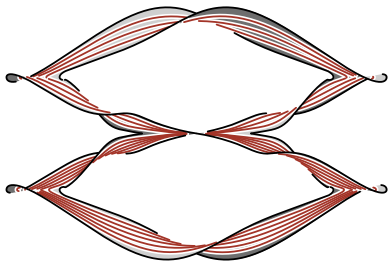
[Ste26]

A Hard-to-Realize Curve

One simple closed curve on $\Sigma_{1,1}$: class $3[A_1] + 5[A_2]$, the same curve in both pictures



abstract page



on the ribbon (smooth, not Legendrian)

A Legendrian realization exists since $[\alpha] \neq 0$ — but what is its front?

LRP: [Kan98; Hon00] page and ribbon as on slide 8 curve drawn for this talk

Stenhede's Algorithm

Setup

Orient each core; pick a 1–handle $H_\star$ whose cocore meets $\mathfrak{a}$ an odd number of times

Bookkeeping

1. **Subdivide**
2. **Relative gain**
3. **Balance total gain**
4. **Prominence P**

Drawing the front

5. **1–handles**
6. **0–handles**
7. **Connect**

[Ste26, §5.2]

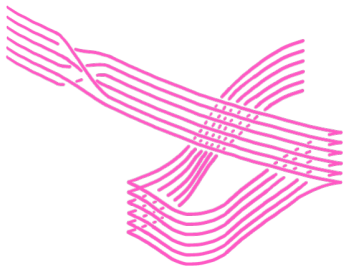
Stenhede's Algorithm: One 1–Handle

Step 5 in a single 1–handle containing six segments of $\mathfrak{a}$



(a) core cut open; six vertical copies [Ste26, Fig. 5.9]

(b) prominences at both ends; braid schematic [Ste26, Fig. 5.11]

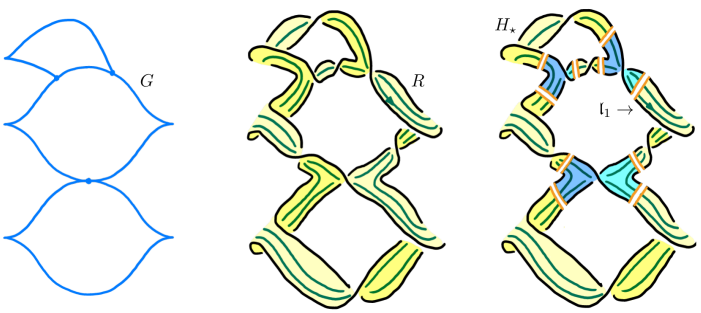


(c) resulting Legendrian braid [Ste26, Fig. 5.12]

Figures reproduced from [Ste26]

Stenhede's Algorithm: A Full Example

Input: a curve on a ribbon $\longrightarrow$ output: its Legendrian realization



graph G; ribbon R with $\mathfrak{a}$ (green); handles and segments, ℓ_1 and $H_\star$ marked [Ste26, Fig. 5.16]

$\longrightarrow$

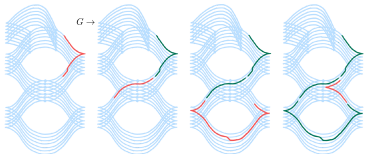


the Legendrian realization of $\mathfrak{a}$ [Ste26, Fig. 5.21]

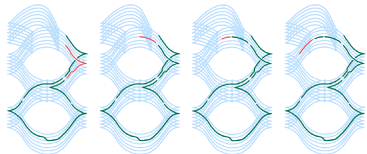
Figures reproduced from [Ste26] [Ste26, §5.3]

A Full Example: Piece by Piece

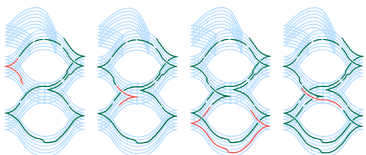
The whole curve, piece by piece, over vertical copies of G (light blue); newest piece in red



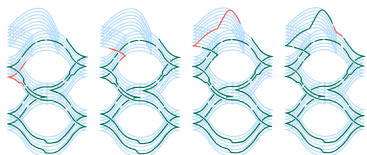
segments ℓ_1–ℓ_4 [Ste26, Fig. 5.17]



segments ℓ_5–ℓ_8 [Ste26, Fig. 5.18]



segments ℓ_9–ℓ_{12} [Ste26, Fig. 5.19]



segments ℓ_{13}–ℓ_{16} [Ste26, Fig. 5.20]

Figures reproduced from [Ste26]

Open Books $\to$ Surgery Diagrams: Avdek vs. Stenhede

	Avdek (2013)	Stenhede (2026)
Page	$\Sigma_{g,1}$ (connected binding)	any genus, any number of boundary components
Twist curves	Lickorish generators	any homologically nontrivial curves
Realization	fixed model curves	explicit algorithm, unique up to Legendrian isotopy

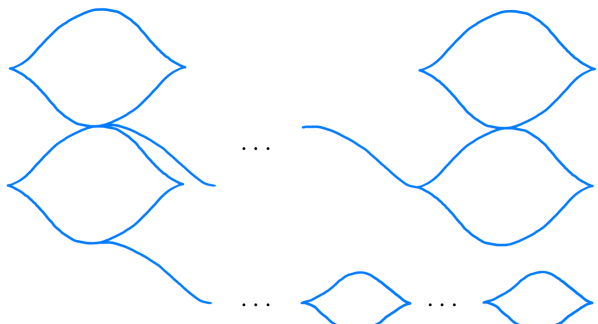
Corollary 6.1 (Stenhede)

φ a product of twists along homologically nontrivial curves $\Rightarrow$ surgery diagram with one component per twist of φ, plus $(+1)$–surgered $\mathrm{tb} = -1$ unknots

[Avd13; Ste26]

Model Pages for Any Surface

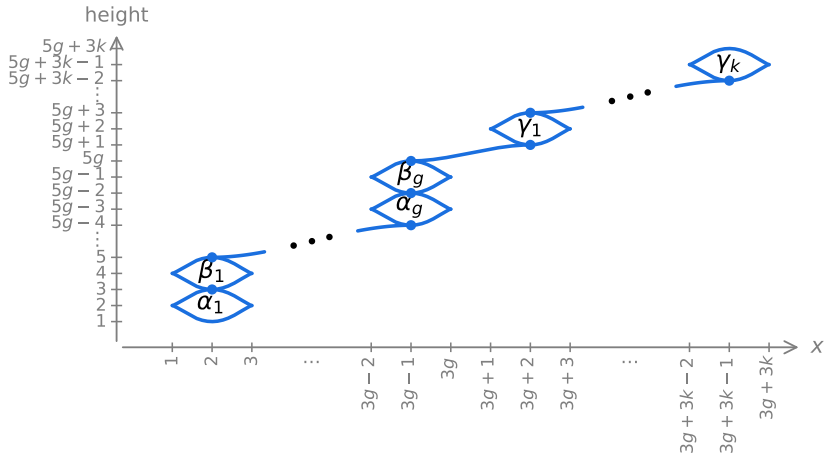
Stenhede's model graphs: every $\Sigma_{g,b}$ is a page of an open book for (S^3, ξ_{std})



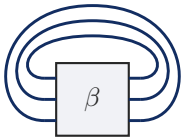
genus = number of A–pieces, boundary components = $1+$ number of B–pieces [Ste26, Fig. 6.2]

Figure reproduced from [Ste26] [Ste26, §6, Thm. 2.5]

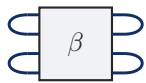
A Computational Complex



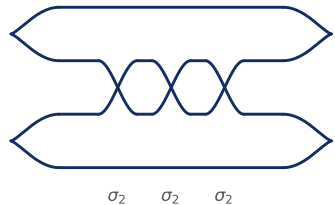
What Is a Plat?



braid closure



plat closure



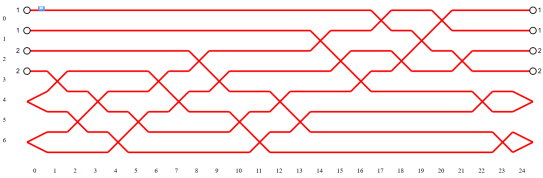
Legendrian plat position

- **Plat**: a braid on $2n$ strands, closed by n caps and n cups
- Every link is a plat; $2n$ strands $=$ an n–bridge presentation [Bir76]
- **Legendrian plat position**: left cusps far left, right cusps far right [Ng03]
- A front becomes a braid word (here σ_2^3 on 4 strands): finite data for a computer

[Bir76; Ng03] closures: standard schematics right: the standard Legendrian trefoil

Application of Plats: Legendrian Contact Homology DGA

Lagrangian resolution of the plat with front crossings 3,5,4,6,5,3,4,2,3,5,6,4,5,1,2,3,0,2,1:



Knots with classical invariants:

Knot label	Color	tb	rot
0		-1	0

Chords are labeled by their x-coordinates:

	1	2	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
From knot	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
To knot	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

No non-trivial augmentations found for LCH

LCH generators	Grading (over $\mathbb{Z}$)	del (over $\mathbb{Z}/2\mathbb{Z}$)
{10}	0	0
{11}	1	1 + {10}*{1}
{12}	1	C0_2_3_1 + C0_2_3_1*{1}*{10}
{13}	2	C0_2_3_1*{1}*{11} + {12}*{1}
{14}	-1	0
{15}	0	{14}
{16}	0	0
{17}	0	C0_0_1_0*{14}
{18}	0	0
{19}	1	C0_0_1_0*{15}*{18} + C0_0_1_0*{16} + {17}*{18}
{1}	0	0
{20}	1	C0_0_1_0 + {18}
{21}	1	C0_0_1_0*{15} + C0_2_3_1 + {17}
{22}	1	1 + {16}*{1}

Sources

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Open books and the Giroux correspondence

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Legendrian realization

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Surgery diagrams and open books

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Plats and fronts

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Figures: slides 17–19 and 21 reproduced from [Ste26]; all other diagrams drawn for this talk, designs credited on each slide